

32524 LANs and Routing (Spring 2023)

Due Date:

IPv4 Addressing Milestone Submission (Tables A, B and C):

By the end of Week 5 (Emailed to your Lab Instructor by **8 am Monday 11 Sept. 2023**)

Video Demonstration Submission:

By the end of Week 10 (Emailed to your Lab Instructor by **8 am Monday 23 Oct. 2023**)

1. Preamble

This Case Study offers students an opportunity to put into practice the network design, implementation, and troubleshooting skills they will have gained through their studies of *LANs and Routing*.

The Case Study is structured to guide groups through the gradual completion of the entire project, enriching their learning journey. The Scenario provides a broad overview of the project, including the rationale behind network construction. The Case Study comprises various tasks, each accompanied by detailed requirements to lead your group through successive steps. It is important to thoroughly read and comprehend each requirement, completing the tasks on a weekly basis to advance your study.

2. Objectives and General Assessment Criteria

Objectives

- Design IPv4 addressing schemes that meet the addressing requirements.
- Configure OSPFv2, including simple route redistribution, for IPv4.
- Configure static, default static, summary static routing, and floating static routing for IPv4.
- Configure switching networks for VLANs, 802.1q trunking, and inter-VLAN routing.
- Configure static NAT and dynamic NAT with overloading.
- Verify the network's functionality and troubleshoot it as needed.

General Assessment Criteria

- Requirements met.
- Correctness of the design.
- Functionality of implementation.
- **Justification** of design and implementation.
- **Verification** of functionality.

3. Assessment

The case study is intended to be completed as a group activity with 75 percent of its total marks being awarded individually. The Case Study's assessment consists of the following two-stage submissions:

1.1 Part A: IPv4 Addressing Milestone Submission (25%; due by the end of **Week 5**)

Each group as a whole is required to Email their IPv4 subnetting and addressing scheme (including Tables A, B, and C on Page 10 of the book) to their lab instructor by the end of Week 5. Feedback on this milestone submission will be provided during the following week's lab class.

1.2 Part B: Packet Tracer based Video Demonstration (75%; individually assessed; due by **Week 10**):

By the end of Week 10, students as groups are required to submit a **recorded video** showing each member orally present their Packet Tracer (PT) network, explaining how each part of their network was designed and implemented to meet the specified requirements.

Additionally, the group must submit a copy of their completed PT file for reference. PS. The marking is based on students' presentation in their submitted video.

For further information about the demonstration, please refer to **Section 7** on **Page 9** of this Case Study book.

4. Suggestions and General Requirements for Case Study Completion

It is expected that each student need to spend approximately six hours to complete this Case Study.

Students as groups are encouraged to plan and work on the Case Study on a weekly basis, aligning with the relevant topics covered in class, as outlined below. This strategy guarantees comprehensive completion of all tasks and ensures students derive the utmost advantage from this task.

Weeks 4&5: complete Task One – IPv4 Addressing (Part A submission due by Week 5)

Week 7: complete Task Two – Static and OSPF Routing

Week 8: complete Task Three – Switching

Week 10: PT Video Demonstration (Part B submission due by Week 10)

Please note, postponing the Case Study until the last week or day(s) could potentially lead to a poorly designed network and restrict the advantages it offers for your final assessments. Hence, we strongly suggest adhering to the recommended weekly plan to secure a successful outcome.

5. Scenario

The objective of this Case Study is to present a scenario in which the *Arcadia Institute of Technology* (AIT), a training organisation, has procured and relocated to a new training centre in the city. This move necessitates a comprehensive network overhaul and implementation. Your group has been engaged as AIT's ICT consultants for this endeavour. Your mission is to design and deploy a new network, a prototype of which will be evaluated through a Packet Tracer-based demonstration.

AIT's campus network encompasses diverse networks, as illustrated in the partial logical Topology Diagram shown in Fig. 1. Your task involves designing and implementing these networks, including a simulated ISP, utilising the interfaces/ports of your preference. AIT expects a network prototype before full deployment to ensure its alignment with their requirements.

To help your group organise the Case Study, it is divided into four tasks, each with specific requirements. Upon accomplishing all tasks, the network prototype is anticipated to be realised using Packet Tracer, followed by a demonstration of its operational capabilities in a recorded video.

Task One: Addressing the Network

The ISP links:

The ISP has allocated **one of the following public IPv4 address spaces for your group**. A '/30 address space' will be used for each of the two ISP links. The remainder addresses will be used for NAT (see Task Four):

- Group A/1: **209.165.200.160/28**
- Group B/2: **209.165.200.176/28**
- Group C/3: **209.165.200.192/28**
- Group D/4: **209.165.200.208/28**
- Group E/5: **209.165.200.224/28**
- Group F/6: **209.165.200.240/28**

The Internal Network:

As part of the network redesign, the AIT has allocated **one of the following private address spaces for your group** for addressing the internal network:

- Group A/1: **172.17.160.0/20**
- Group B/2: **172.17.176.0/20**
- Group C/3: **172.17.192.0/20**
- Group D/4: **172.17.208.0/20**
- Group E/5: **172.17.224.0/20**
- Group F/6: **172.17.240.0/20**

The expected numbers of users for each of the user groups are:

For the **West** switching network (this will be implemented in VLANs):

- 1000 hosts for Students
- 120 hosts for Staff

On the **Main** site:

- 30 hosts on the Server Farm LAN.

On the **Branch** site:

- 250 hosts in Students (as shown in Appendix: Partial Topology Diagram)
- 500 hosts in a spare LAN

The AIT requires that

- ☐ The use of **VLSM** design to maximise the use of IPv4 addresses.
- ☐ **All networking devices (including switch SVIs) must have IPv4 addresses** and the PC hosts' **gateways** will use the **first** usable address(es) in their subnet.
- ☐ Each of the ISP links will be allocated a **subnet mask of /30**.
- ☐ The **Management VLAN** for the switching networks will have **an extra host** (referred to as the "Network Admin Host" as shown in the Topology Diagram) for network administration usage.

At this stage, AIT agrees that it is sufficient to assign all hosts with an IPv4 address statically.

Milestone Submission: Tables A, B and C

- 1) IPv4 Network subnetting **Table A**, which shows possible subnets that meet the design requirements; Subnets that are not used are to be clearly identified in each table ('not in use').
- 2) Detailed IPv4 addressing tables (**Tables B and C**) showing all networking devices' names and their interface details. Note that, the gateways of the VLANs will be implemented as *sub-interfaces*, e.g., Fa0/0.10 as the gateway of VLAN10.

Discussion Questions: Consider how you do subnetting so as to meet each of the requirements.

Task Three: Switching Network at City and Branch Campuses

Because of the size and complexity of LANs, AIT wants to use **VLAN** technologies to control broadcasts, enhance security and logically organise its user groups. 802.1Q trunk-based Inter-VLAN routing is to be implemented to advertise all VLAN networks.

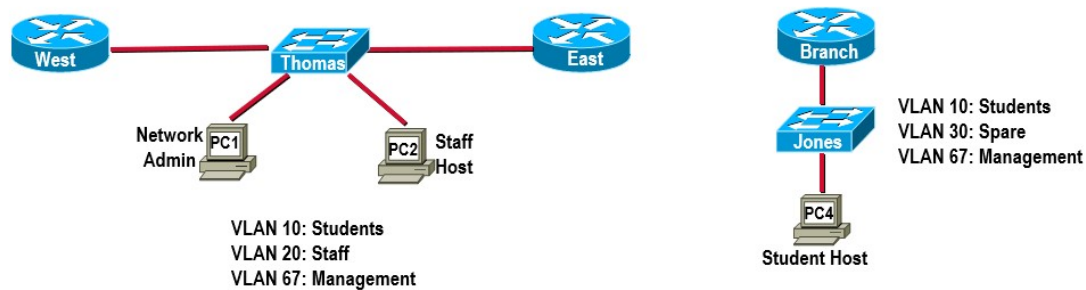


Fig. 3. The switching networks at the City Campus (left) and the Branch Campus (right).

The switching networks at the City and Branch Campus:

- Create VLANs with the following VLAN IDs and Names for the required networks:
 - VLAN 10 – Students
 - VLAN 20 – Staff
- For the purpose of demonstrating network functionality, allocate **five** ports to each user VLAN on each switch.
- Create **VLAN 67** as the Management VLAN and allocate **one** port to this VLAN on each switch. Use **VLAN 555** as the native VLAN ID.
- Do NOT allow traffic of the default VLAN 1 and unknown VLANs onto the trunk link(s).

Discussion Questions:

Explain, in detail, how you **implement** (e.g., show the corresponding commands used from the device's running-configure outputs) and **use specific tools/commands to verify** the following functions:

- 1) Creating VLANs and assigning ports.
- 2) SVI configuration and verification.
- 3) Inter-VLAN routing configuration and verification.

Task Four: IP Addressing Services

The Company has been allocated one of the following blocks of public IPv4 addresses for the ISP links with the remainder ones used for the IPv4 NAT pools:

- Group A/1: **209.165.200.160/28**
- Group B/2: **209.165.200.176/28**
- Group C/3: **209.165. 200.192/28**
- Group D/4: **209.165. 200.208/28**
- Group E/5: **209.165. 200.224/28**
- Group F/6: **209.165. 200.240/28**

Requirements on NAT for IPv4:

- ☐ All devices and only these devices in the internal network are expected to have Internet connectivity using the available addresses from the public address pool with overloading.
- ☐ For demonstration purposes, define a **static NAT** for the **Serve host (PC4)** using an available address from the pool(s).
- ☐ Adjust the **static routing configuration** on the ISP so that returning traffic from the Internet and external services can return to the internal network.

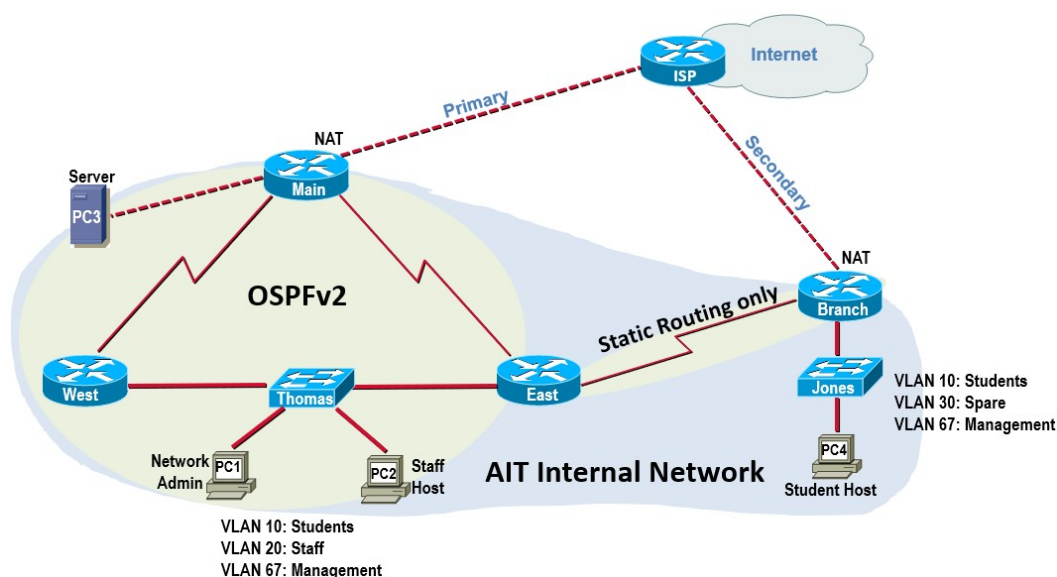


Fig. 4. IP Addressing Services

Discussion Questions:

- 1) Show the details of your design, such as NAT pools for IPv4, and partial configuration scripts specific to this task that help to justify your solution.
- 2) Explain how the static routing over the two ISP links has changed with the implementation of NAT.
- 3) Discuss how you verify that the functionality of your design meets all requirements.

7. Packet Tracer based Video Demonstration

The Company now wants a demonstration of the prototype network. To do this, you need to set up the network that you have designed and configure devices in Packet Tracer to demonstrate all the required functions.

The demonstration requires **basic settings on all routers and switches** including hostname, passwords, MOTD banner, management address and **SSH** access, detailed as follows:

- ☐ Configure basic device settings for routers and switches:
 - Configure **hostnames** as per the partial Topology Diagram.
 - Configure password **cisco** for console connections.
 - Encrypt the privileged EXEC mode using the password **class**.
 - Disable Domain Name Server **(DNS) lookup**.
 - Enable **logging synchronous** for console connections and all virtual terminal lines.
- ☐ Configure the interfaces of routers and hosts as per the Topology Diagram and your Addressing Tables B&C.
- ☐ Configure static and OSPF routing on all routers according to the requirement and your design.
- ☐ Configure VLANs according to your Switch and VLAN tables.
- ☐ Configure the Management VLAN SVI interface on the switches.
- ☐ Configure the following host PCs based on your addressing:
 - **PC1** as the Network Admin Host in Management VLAN on **Thomas**.
 - **PC2** as the Staff Host on **Thomas**.
 - **PC3** as the Server connected to the **Main** router.
 - **PC4** as a Student Host on **Jones**.

PS. The Company has a plan to implement DHCPv4 service and use ACLs to control network traffic. At this stage, statically assigning IPv4 addresses is sufficient for the demonstration purpose.

The Company requires the following network verification to be assessed:

- Verification of the devices' basic configuration.
- Verification of the correctness and functionality of the interface configuration.
- Verification of dynamic routing for IPv4.
- Verification of static routing for IPv4.
- Verification of the VLANs and inter-VLAN routing.
- Verification of NAT (including Overloaded NAT and static NAT).
- Verifying end-to-end connectivity of all hosts to each other and the ISP's loopback addresses.
- Verification of the redundant links.

Discussion Questions:

- 1) **Discuss how to verify each of the above functions (the commands, expected outcomes, and explanation on the device outputs).**

Sample Partial Tables:**Table A - IPv4 Subnetting Table**

Subnet Number*	Subnet Address	Subnet Mask	Hosts Required	Maximum Hosts in Subnet	In Use (Yes or No)	Network Name

*"Subnet Number" is the index (starting from 0) of the subnet as the result of subnetting.

Table B Device Interface IP Addressing Table

Device	Interface	IPv4 address	Subnet Mask
ISP	Fa0/0		
	Fa0/0		
West			
...			
Thomas	SVI		
...			

Table C Host Addressing Table

Host	IPv4 Address	Subnet Mask	Gateway
PC1			
PC2			
PC3			
PC4			

Table D Switch Table

Switch Name:

Switch Management IP Address:

Interface type & Port Number	Description of Purpose	Port Bandwidth	Network Name	Subnet Address	Subnet Mask	VLAN ID & Name	Switch Port Mode	Layer 2 Encapsulation

Table E VLAN Table

Switch Name	Number of Ports	Location	IP Address	Gateway	VLAN ID & Name